

Problem 1. Let G be a group. For every element $g \in G$, one considers the function

$$\text{mult}_g : G \rightarrow G \quad x \mapsto g \cdot x.$$

- (i) Let $e \in G$ denote the identity element. What is mult_e ?

Solution. $\text{mult}_e = \text{id}_G$ is the identity function for G .

- (ii) Show for all $g, g' \in G$ that

$$\text{mult}_g \circ \text{mult}_{g'} = \text{mult}_{g \cdot g'}.$$

Solution. For all $x \in G$ we have by associativity that

$$\begin{aligned} (\text{mult}_g \circ \text{mult}_{g'})(x) &= \text{mult}_g(\text{mult}_{g'}(x)) \\ &= g \cdot (g' \cdot x) = (g \cdot g') \cdot x = \text{mult}_{g \cdot g'}(x). \end{aligned}$$

- (iii) Show for all $g \in G$ that the function mult_g is bijective.

Solution. Every element $g \in G$ has an inverse $g^{-1} \in G$ satisfying

$$g \cdot g^{-1} = e = g^{-1} \cdot g.$$

Applying (i) and (ii) shows that

$$\text{mult}_g \circ \text{mult}_{g^{-1}} = \text{id}_G = \text{mult}_{g^{-1}} \circ \text{mult}_g,$$

i.e. mult_g is a bijection with inverse $\text{mult}_{g^{-1}}$.

(iv) Show that the function

$$\mu : G \rightarrow \text{Sym}(G) \quad g \mapsto \text{mult}_g$$

is an injective group homomorphism.

Solution. We know from (ii) that μ is a group homomorphism. For injectivity, we observe that $\text{mult}_g = \text{mult}_{g'}$ implies

$$g = g \cdot e = \text{mult}_g(e) = \text{mult}_{g'}(e) = g' \cdot e = g'.$$

Problem 2. Let $H \subset G$ be a subgroup of a group.

- (i) In class we defined the set G/H to be the quotient $G/\sim_H$ of G by a certain equivalence relation $\sim_H$ on G . Give the definition of $\sim_H$.

Solution. By definition $g \sim_H g'$ if and only if $g' = g \cdot h$ for some $h \in H$.

- (ii) Define the *left coset* gH of an element $g \in G$. State the relation between gH and the equivalence class of g with respect to $\sim_H$.

Solution. In Lecture XIII we defined

$$gH = \{g \cdot h : h \in H\}.$$

and proved that gH is equal to the equivalence class of g with respect to $\sim_H$.

(iii) Let $g \in G$ be an element. Construct a bijection between gH and H .

Solution. We claim that the function

$$H \rightarrow gH \quad h \mapsto g \cdot h$$

is bijective. It is tautologically surjective, and it is injective by 1(iii).

(iv) Suppose now that G is finite. Deduce that

$$|G| = |H| \cdot |G/H|.$$

Solution. According to (ii), G is the disjoint union of its H -cosets, and the number of such cosets is $|G/H|$. According to (iii), each coset has cardinality $|H|$.

(This is Lagrange's theorem from Lecture XIII.)

Problem 3. Let $N \subset G$ be a subgroup of a group.

- (i) What does it mean to say that N is a *normal subgroup*?

Solution. We say that N is a normal subgroup of G if $gN = Ng$ for all $g \in G$.

- (ii) Suppose for the remainder of this problem that N is a normal subgroup. Show that there is a binary operation

$$G/N \times G/N \rightarrow G/N \quad (x, y) \mapsto x \cdot y$$

with the property that

$$[g] \cdot [g'] = [g \cdot g'] \quad \text{for all } g, g' \in G. \quad (*)$$

Solution. We need to verify the implication

$$g_1 \sim_N g'_1 \quad \text{and} \quad g_2 \sim_N g'_2 \quad \Rightarrow \quad g_1 \cdot g_2 \sim_N g'_1 \cdot g'_2$$

For $i = 1, 2$, choose $n_i \in N$ so that $g'_i = g_i \cdot n_i$. Then

$$g'_1 \cdot g'_2 = g_1 \cdot n_1 \cdot g_2 \cdot n_2.$$

Now $Ng_2 = g_2N$ because N is a normal subgroup, so $n_1 \cdot g_2 = g_2 \cdot n'_1$ for some $n'_1 \in N$. Therefore,

$$g'_1 \cdot g'_2 = g_1 \cdot g_2 \cdot n'_1 \cdot n_2.$$

Finally, $n'_1 \cdot n_2 \in N$ because subgroups are closed under the group operation, so the proof is complete.

- (iii) Show that this binary operation makes G/N into a group, and that the quotient map $\pi : G \rightarrow G/N$ is a group homomorphism.

Solution. The binary operation is associative because

$$\begin{aligned} ([g] \cdot [g']) \cdot [g''] &= [g \cdot g'] \cdot [g''] = [(g \cdot g') \cdot g''] \\ &= [g \cdot (g' \cdot g'')] = [g] \cdot [g' \cdot g''] = [g] \cdot ([g'] \cdot [g'']) \end{aligned}$$

for all $g, g', g'' \in G$. The element $[e] \in G/N$ is an identity element because

$$[e] \cdot [g] = [e \cdot g] = [g] = [g \cdot e] = [g] \cdot [e]$$

for all $g \in G$. Finally, for all $g \in G$, $[g^{-1}]$ is an inverse to $[g]$ because

$$[g] \cdot [g^{-1}] = [g \cdot g^{-1}] = [e] = [g^{-1} \cdot g] = [g^{-1}] \cdot [g].$$

The equation (*) says exactly that π is a group homomorphism.

- (iv) Let $N \subset G$ be a normal subgroup, and let $\phi : G \rightarrow G'$ be a group homomorphism. State the criterion for there to exist a homomorphism

$$\psi : G/N \rightarrow G'$$

satisfying $\psi \circ \pi = \phi$.

Solution. Such a homomorphism exists if and only if $N \subset \ker(\phi)$.

Problem 4. Let G and G' be finite groups. Suppose that the order of G and the order of G' are coprime. Show that every homomorphism $\phi : G \rightarrow G'$ is trivial.

Solution. Since the image of a homomorphism is a subgroup, Problem 2(iv) implies that the order of $\text{im}(\phi)$ divides the order of G' . Since $\text{im}(\phi) \simeq G / \ker(\phi)$, it also implies that the order of $\text{im}(\phi)$ divides the order of G . Since the orders of G and G' are coprime, this implies that $\text{im}(\phi)$ has order 1, i.e. that it is the trivial subgroup of G' . This means that ϕ is the trivial homomorphism.

Problem 5. This problem is about the $(7,4)$ Hamming code constructed from the generator polynomial

$$g(X) = X^3 + X^2 + 1.$$

- (i) Compile the Zech logarithm table for $\mathbf{F}_2[X]/g\mathbf{F}_2[X]$ with base X .

Solution. The logarithm table is:

X^n	n	X^n	n
0	—	$X^2 + 1$	3
1	0	$X^2 + X + 1$	4
X	1	$X + 1$	5
X^2	2	$X^2 + X$	6

(ii) Determine whether or not the message

$$X^5 + X^4 + X^3 + X^2 + X + 1$$

contains any errors. If so, correct the error and decode the message.

Solution. We compute that

$$\begin{aligned} X^5 + X^4 + X^3 + X^2 + X + 1 \\ &= (X + 1) + (X^2 + X + 1) + (X^2 + 1) + X^2 + X + 1 \\ &= X^2 + X = X^6. \end{aligned}$$

Therefore, the corrected message is

$$X^6 + X^5 + X^4 + X^3 + X^2 + X + 1 = (X^3 + X + 1) \cdot g(X)$$

and the decrypted message is

$$X^3 + X + 1.$$